

Can exercise-related improvements in immunity influence cancer prevention and prognosis in the elderly?

Cancer incidence increases with advancing age. Over 60% of new cancers and 70% of cancer deaths occur in individuals aged 65 years or older. One factor that may contribute to this is immunosenescence - a canopy term that is used to describe age-related declines in the normal functioning of the immune system. There are multiple age-related deficits in both the innate and adaptive systems that may play a role in the increased incidence of cancer. These include decreased NK-cell function, impaired antigen uptake and presentation by monocytes and dendritic cells, an increase in 'inflammaging', a decline in the number of naïve T-cells able to respond to evolving tumor cells, and an increase in functionally exhausted senescent cells. There is consensus that habitual physical exercise can offer protection against certain types of cancer; however the evidence linking immunological mechanisms, exercise, and reduced cancer risk remain tentative. Multiple studies published over the last two decades suggest that exercise can mitigate the deleterious effects of age on immune function, thus increasing anti-cancer immunity. The potential ameliorative effect of exercise on these mechanisms include evidence that physical activity is able to stimulate greater NK-cell activity, enhance antigen-presentation, reduce inflammation, and prevent senescent cell accumulation in the elderly. Here we discuss the role played by the immune system in preventing and controlling cancer and how aging may retard these anti-cancer mechanisms. We also propose a pathway by which exercise-induced alterations in immunosenescence may decrease the incidence of cancer and help improve prognosis in cancer patients. Copyright © 2013 Elsevier Ireland Ltd. All rights reserved.